Abstract

Background: Law enforcement resource allocation is being more and more influenced by crime prediction models that are also prone to reinforce biases towards the vulnerable groups.
Objective: This paper presents the Fairness-Constrained Multi-Task Long Short-Term Memory (FC-MT-LSTM) architecture that is clearly optimized in terms of prediction accuracy and fairness amongst the protected groups.
Materials and Methods: Against this extensive crime data, we compare our method with six baseline models, namely, SARIMA, Prophet, Random Forest, XGBoost, CNN-LSTM, and Transformer, using national crime data of 36 states and 700+ districts in India (National Crime Records Bureau, 2017–2022). This dataset involves 21,067 records of crime in four areas of protection namely; Scheduled Castes (SC), Scheduled Tribes (ST), women and children with 39 to 52 different types of crime per category. Our FC-MT-LSTM system integrates spatial convolutional networks, temporal bidirectional LSTMs with attention, a common encoder, and particular group decoders (with access to fairness-constrained loss).
Results: Experimental findings indicate that FC-MT-LSTM has a balance in performance across all the groups that are being protected (Fairness Ratio: 1.99) and competitive accuracy (MAE: 6.54, R^2: 0.9922) when compared to the performance of ensemble methods with fairness disparities of 3.7–4.7$\times$ in spite of high aggregate accuracy.
Conclusion: This work contributes a deployable framework for equitable crime forecasting with implementations in both PyTorch and TensorFlow.

***Index Terms*—Algorithmic fairness, crime prediction, multi-task learning, LSTM, deep learning, protected groups, bias mitigation**

I. Introduction

The criminal justice system is using a wide range of algorithms to make decisions about how to plan patrols, use resources, and prevent crime [1]. Machine learning algorithms can make very accurate predictions about future crime by looking at past crime data, socioeconomic factors, and patterns over time. However, these approaches exhibit a significant flaw: historical crime data consistently reveals structural biases in police operations, crime reporting, and societal inequalities [2], [3].

When machine learning models are trained without fairness constraints on biased data, they increase the risk of aggravating existing disparities by directing more police resources to communities that are already over-policed, but not enough to help vulnerable groups [5], [6]. Police resources go to communities that are already over-policed. Standard crime prediction models use only accuracy metrics like Mean Absolute Error (MAE), R-squared values, or Root Mean Square Error (RMSE). This single-objective optimization does not take impartiality into any account, which could lead to models to make very different predictions for secured groups that do well overall. In particular:

- Models may consistently undervalue or overvalue crimes affecting women and children.
- The accuracy of predictions varies a lot between different demographic groups.
- Giving out resources based on wrong projections keeps old wrongs going. Why not model groups separately? It doesn't show how crime trends are different for vulnerable groups.

The goal is to make a crime prediction system that is fair for all protected groups and very accurate.

A. Research Objectives

The objectives of this study are to examine the trade-off between fairness and accuracy in crime prediction:

1) Build a deep learning architecture that can do more than one thing and clearly shows crime patterns for four protected groups: SC, ST, Women, and Children.
2) Use fairness constraints to make sure that predictions disparities are minimum across vulnerable populations during model training .
3) Use real-world crime data to see how fairness standards stack up against known accuracy standards.
4) Look at the suggested FC-MT-LSTM model next to six of the best baseline models.
5) Show how it could be used in the real world to fairly divide up police resources

B. Contributions

Here are the most important things our work adds:

- **New Architecture**: We introduce FC-MT-LSTM, which uses spatial CNNs, temporal attention-based LSTMs, shared encoding, and group-specific decoders with fair loss functions.
- **Extensive Benchmarking**: We use six different baselines, such as statistical (SARIMA and Prophet), ensemble (Random Forest and XGBoost), and deep learning (CNNLSTM and Transformer) methods.
- **Full Fairness Evaluation**: We make metrics that look at the differences in predictions between SC, ST, Women, and Children groups and are useful for each group's weaknesses.
- **Real-World Dataset**: From India's NCRB, we use six years of expanded crime data. This data has protected category annotations, which cover more than 700 districts.
- **Practical Framework**: We leveraged Python code in both PyTorch and TensorFlow.

II. Related Work

A. Crime Prediction Models

Crime forecasting has changed from traditional statistical methods to advanced deep learning models. Initial work depended on time-series models like ARIMA and its variations to modelcrime patterns over time [7].More recent methods include spatial information using geographically weighted regression [8]. The use of machine learning guided to important

improvements in accuracy. Random Forest and XGBoost have shown useful results on tabular crime data because they capture non-linear relationships between economic and social factors and criminal activity [9]. Prophet, created by Facebook, offers solid projection for data with clear periodic trends [10].

In recent years, deep learning architectures have recorded state-of-the art performances. Convolutional Neural Networks (CNNs) extract locational characteristics of heatmaps of crime [11]. Mechanisms of attention determine key time intervals, whereas Long Short Term Memory networks represent crime sequence temporal dependencies [12]. Transformer architectures have been adapted with crime prediction with promising results [13]. Despite such enhancements, the majority of crime prediction studies are interested in maximizing the accuracy only and do not consider the aspects of fairness among the groups that are being protected.

B. Algorithmic Fairness in Criminal Justice

Following the notable examples of discriminative risk assessment instruments, the issue of the fairness of the algorithmic decision-making in criminal justice has gained more attention. The COMPAS recidivism prediction system was discovered to have a high degree of racial disparities [5], [14].

Various sets of researchers have provided their distinct approach to measurement of fairness. One rule is known as demographic parity which states that each group should be given the same ratio of positive predictions. The second rule is equalized odds, which states that the true and the false rate should be equal in every group. The third rule, predictive parity requires that the probability that a prediction will be correct once it is called positive should be identical across all groups [15]. Nonetheless, results on impossibility in mathematics demonstrate that one cannot have these notions met simultaneously except in the trivial sense [16].

There are three categories of fairness interventions - the methods that work prior to training modify the data that is being used to teach the model. The methods applied during training introduce fairness regulations in the learning process. After training approaches modify the scores that the model generates [19]. Our FC-MT-LSTM utilizes in processing approach by penalizing fairness violation in the loss directly.

C. Multi-Task Learning for Crime Prediction

Multi-task learning (MTL) uses common representations between related tasks in order to enhance generalization and sample efficiency [20]. MTL has the ability to model various types of crimes or geographic areas at the same time in predicting crime [21].

Our FC-MT-LSTM is an expansion of the traditional MTL in that task definition is done on the basis of group protection as opposed to crime type. The benefit of this centric formulation is that it is able to optimize fairness explicitly within vulnerable groups. The common encoder is the one that learns common crime patterns and group-specific decoders individualize characteristics that influence SC, ST, Women, and Children.

D. Summary of Existing Approaches

Table I provides a comprehensive comparison of crime prediction methodologies, highlighting their techniques, strengths, and limitations.

The literature reveals three key gaps that motivate our work: (1) most crime prediction models optimize solely for accuracy without fairness considerations; (2) fairness research in criminal justice focuses primarily on recidivism and risk assessment rather than crime forecasting; and (3) existing fairness-aware approaches do not leverage multi-task learning for group-specific optimization.

III. Data Description and Preprocessing

A. Dataset Overview

We use the National Crime Records Bureau (NCRB) of India's crime data from 2017 to 2022. There are 36 states and union territories in the dataset, as well as 700 districts. There are a total of 21,067 records, which are divided into four protected groups. Table II gives a quick overview of the dataset's features. Table II summarizes the dataset characteristics.

B. Protected Group Definitions

We focus on four protected groups that face systemic vulnerabilities in the Indian criminal justice context:

1) **Scheduled Castes (SC)**: Communities historically subjected to caste-based discrimination and social exclusion
2) **Scheduled Tribes (ST)**: Indigenous tribal populations facing geographic isolation and limited access to justice
3) **Women**: Gender-based crimes including domestic violence, sexual assault, and harassment
4) **Children**: Minors vulnerable to trafficking, sexual abuse, and exploitation under POCSO (Protection of Children from Sexual Offences) Act

Table III shows the distribution across protected groups.

C. Feature Engineering

We construct 188 features across multiple categories:

Temporal Features: Time-lagged crime counts (1–3 year lags), rolling statistics (mean, standard deviation over 2–3 year windows), year-over-year change percentages, and temporal encodings (year position, normalized year).

Spatial Features: District-level crime aggregates, state-level statistics, crime density metrics, and district/state crime rankings.

Crime Category Features: Counts for specific crime types (violent crimes, sexual crimes, property crimes, kidnapping), with category-specific aggregations.

Group Features: Protected group indicators (one-hot encoding), group proportion within districts, and group-specific crime rates.

TABLE I
LITERATURE SURVEY: CRIME PREDICTION MODELS AND FAIRNESS APPROACHES

Study/Model	Methodology	Key Contributions	Limitations
Statistical Time-Series Methods			
[7]	Autoregressive integrated moving average with seasonal components	Captures temporal autocorrelation; handles trend and seasonality; interpretable parameters	Assumes linear relationships; requires stationary data; no spatial modeling; poor with irregular patterns
[10]	Decomposable additive model with trend, seasonality, holidays	Robust to missing data; automatic changepoint detection; handles multiple seasonalities	Limited feature incorporation; struggles with complex non-linear patterns; no fairness consideration
Ensemble Machine Learning Methods			
[9]	Ensemble of decision trees with bootstrap aggregation	Handles non-linear relationships; feature importance ranking; resistant to overfitting	Black-box nature; memory intensive; no temporal sequence modeling; fairness not addressed
[22]	Gradient boosted decision trees with regularization	State-of-the-art tabular performance; handles missing values; efficient implementation	Requires careful hyperparameter tuning; sequential training; no inherent fairness mechanism
Deep Learning Approaches			
[12]	Recurrent neural network with gated memory cells	Models long-term temporal dependencies; handles variable-length sequences	Computationally expensive; prone to overfitting on small datasets; no spatial awareness
[11]	Hierarchical RNN with attention for crime prediction	Captures hierarchical spatial-temporal patterns; attention for interpretability	Requires large datasets; no explicit fairness constraints; complex architecture
[13]	Self-attention mechanism for sequence modeling	Parallel computation; captures long-range dependencies; state-of-the-art NLP performance	High memory requirements; needs positional encoding; limited inductive bias for time-series
Fairness-Aware Approaches			
[5]	Investigative analysis of recidivism prediction	Exposed racial bias in criminal justice algorithms; sparked fairness research	Analysis only; no proposed solution; focused on recidivism not crime prediction
[18]	Convex optimization with fairness constraints	In-processing fairness; theoretical guarantees; applicable to various classifiers	Limited to convex problems; binary classification focus; accuracy-fairness trade-off
[4]	Under-reporting aware crime prediction	Addresses reporting bias; improves fairness for minority areas	Requires under-reporting estimates; limited to spatial fairness; no multi-task learning
FC-MT-LSTM (Ours)	Multi-task LSTM with fairness-constrained loss	Group-specific decoders; explicit fairness optimization; spatial-temporal fusion	Requires group labels; computational overhead; fairness-accuracy trade-off

TABLE II
DATASET STATISTICS

Attribute	Value
Time Period	2017–2022 (6 years)
States/Union Territories	36
Districts	$\sim$700
Total Records	21,067
Training Period	2017–2021
Test Period	2022
Features Engineered	188

TABLE III
DATASET DISTRIBUTION BY PROTECTED GROUP

Group	Records	%	Crime Categories
Children	5,321	25.3%	52
Women	5,322	25.3%	43
SC	5,323	25.3%	39
ST	5,101	24.2%	39
Total	**21,067**	**100%**	**173 unique**

D. Data Preprocessing Pipeline

The preprocessing pipeline consists of:

1) **Missing Value Handling**: Crime categories with zero reported incidents are preserved as zeros. Exculsion of Missing district or group identifiers (<0.1% of records).
2) **Temporal Split**:Temporal data split —2017–2021 for training (17,379 records) and 2022 for testing (3,688 records).
3) **Feature Scaling**: StandardScaler fitted on training data creates and checks zero mean and unit variance.
4) **Sequence Construction**: For sequential models, historical data was used to create time-series to predict current year's crime counts.

IV. Proposed FC-MT-LSTM Architecture

A. Architecture Overview

The Fairness-Constrained Multi-Task LSTM (FC-MT-LSTM) consists of four primary components:

1) **Spatial CNN**: Extracts crime patterns of geographic and district-level
2) **Temporal LSTM with Attention**: Models dependencies of time series data with adjustable focus on critical periods
3) **Shared Encoder**: Fuses spatial and temporal representations into unified crime embeddings
4) **Group-Specific Decoders**: Four parallel decoders tailored for SC, ST, Women, and Children

Fig. 1 illustrates the complete architecture.

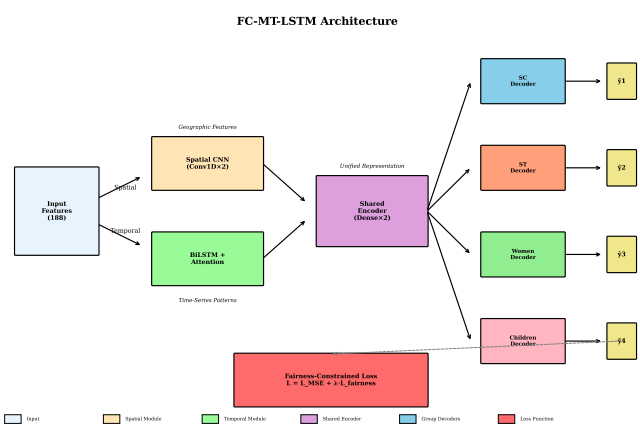

Fig. 1. FC-MT-LSTM Architecture. The proposed architecture uses a CNN to extract spatial features and a bidirectional attention-based LSTM to learn time-dependent patterns. Outputs from these components are combined through a shared encoder, followed by four dedicated decoders for SC, ST, Women, and Children. The objective function incorporates fairness constraints so that both accuracy and group equity are optimized together.

B. Spatial CNN Module

The Spatial CNN captures district-level crime characteristics through convolutional layers:

$$X^{(1)}_{\text{spatial}} = \text{ReLU}(\text{BN}(\text{Conv1D}(X_{\text{spatial}}))) \tag{1}$$

$$X^{(2)}_{\text{spatial}} = \text{ReLU}(\text{BN}(\text{Conv1D}(X^{(1)}_{\text{spatial}}))) \tag{2}$$

$$H_{\text{spatial}} = \text{Dense}(\text{GlobalMaxPool}(X^{(2)}_{\text{spatial}})) \tag{3}$$

To find the local spatial patterns, two 1D convolutional layers with 32 and 64 filters are used, each with a kernel size of 3. Batch normalization and ReLU activation make sure that training is stable. A $d_s = 64$ dimensional spatial representation are the aggregate feature influenced by Global max pooling.

C. Temporal LSTM with Attention

The temporal module employs bidirectional LSTM with additive attention:

$$H_{\text{lstm}} = \text{BiLSTM}(X_{\text{temporal}}) \in \mathbb{R}^{B\times T\times 2d_h} \tag{4}$$

where T is sequence length and $d_h = 128$ is the hidden dimension.

The attention mechanism identifies important timesteps:

$$e_t = \tanh(W_a H_{\text{lstm}}[:,t,:] + b_a) \tag{5}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{T}\exp(e_j)} \tag{6}$$

$$H_{\text{temporal}} = \sum_{t=1}^{T} \alpha_t H_{\text{lstm}}[:,t,:] \tag{7}$$

Attention weights α_t quantify each year's relevance to future crime prediction, enabling interpretability.

D. Shared Encoder

The shared encoder fuses spatial and temporal representations:

$$H_{\text{concat}} = [H_{\text{spatial}}; H_{\text{temporal}}] \tag{8}$$

$$H^{(1)}_{\text{shared}} = \text{Dropout}(\text{ReLU}(\text{BN}(\text{Dense}(H_{\text{concat}})))) \tag{9}$$

$$H^{(2)}_{\text{shared}} = \text{Dropout}(\text{ReLU}(\text{BN}(\text{Dense}(H^{(1)}_{\text{shared}})))) \tag{10}$$

Two fully-connected layers with 256 units transform concatenated features into a unified 256-dimensional crime representation.

E. Group-Specific Decoders

Four parallel decoders generate predictions for each protected group:

$$\hat{y}_g = \text{Decoder}_g(H_{\text{shared}}) \quad \text{for } g \in \{\text{SC, ST, Women, Children}\} \tag{11}$$

Each decoder consists of two dense layers (128 and 64 units) with batch normalization, ReLU activation, and dropout (rate 0.3), followed by a single-unit output layer.

F. Fairness-Constrained Loss Function

The total loss combines prediction accuracy and fairness:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{fairness}} \tag{12}$$

where λ is the fairness penalty weight (we set $\lambda = 0.5$).

Prediction Loss: Mean Squared Error measures prediction accuracy:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B} \sum_{i=1}^{B} (y_i - \hat{y}_{g_i})^2 \tag{13}$$

where g_i is the protected group label for sample i.

Fairness Loss: The fairness penalty minimizes prediction disparity across groups:

$$\mathcal{L}_{\text{fairness}} = \frac{1}{N_{\text{pairs}}} \sum_{g_1 < g_2} |\text{MAE}_{g_1} - \text{MAE}_{g_2}| \tag{14}$$

where MAE_g is the Mean Absolute Error for group g:

$$\text{MAE}_g = \frac{1}{|B_g|} \sum_{i \in B_g} |y_i - \hat{y}_i| \tag{15}$$

This pairwise comparison penalizes unequal accuracy across the four protected groups, encouraging balanced performance.

G. Training Procedure

We train FC-MT-LSTM using:

- **Optimizer**: Adam with learning rate $\eta = 0.001$, weight decay 10^{-5}
- **Batch size**: 32 samples
- **Epochs**: 100 with early stopping (patience = 10)
- **Learning rate schedule**: StepLR with $\gamma = 0.5$ every 20 epochs
- **Gradient clipping**: Max norm 1.0
- **Validation split**: 20% of training data

V. Experimental Setup

A. Baseline Models

We benchmark FC-MT-LSTM against six diverse approaches:

SARIMA: Seasonal ARIMA(1,1,1)(1,1,1,12) models fitted per district-group combination.

Prophet: Facebook's decomposition model with yearly seasonality, changepoint prior scale 0.05.

Random Forest: 100 decision trees, max depth 10, min samples split 5.

XGBoost: 200 trees, max depth 6, learning rate 0.1.

CNN-LSTM: 1D CNN (32 filters) followed by 2-layer LSTM (64 hidden units).

Transformer: 2 encoder layers, 4 attention heads, model dimension 64.

B. Evaluation Metrics

Accuracy Metrics:

- Mean Absolute Error (MAE): $\frac{1}{n} \sum |y_i - \hat{y}_i|$
- Root Mean Square Error (RMSE): $\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$
- Coefficient of Determination (R^2): $1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

Fairness Metrics:

- Fairness Gap: $\max_g \text{MAE}_g - \min_g \text{MAE}_g$
- Fairness Ratio: $\frac{\max_g \text{MAE}_g}{\min_g \text{MAE}_g}$

Lower Fairness Gap and Fairness Ratio (closer to 1.0) indicate more equitable performance across protected groups.

VI. Results

A. Overall Performance Comparison

Table IV presents the comprehensive comparison across all models.

Key Observations:

- **Ensemble methods** (Random Forest, XGBoost) achieve excellent accuracy (MAE < 2.5, $R^2 > 0.999$) but exhibit severe fairness violations (Fairness Ratio: 3.73–4.74).
- **Statistical methods** (SARIMA, Prophet) demonstrate poor accuracy (MAE 135–167) with moderate fairness ratios (1.17–1.22).
- **FC-MT-LSTM** achieves balanced trade-off: competitive accuracy (MAE: 6.54, R^2: 0.9922) with substantially improved fairness (Ratio: 1.99).

The fairness ratio of 1.99 indicates that the worst-performing group experiences MAE only 99% higher than the best-performing group—a substantial improvement over ensemble methods' 273–374% disparities.

Fig. 2 visualizes the performance comparison across all metrics.

B. Per-Group Performance Analysis

Table V presents the per-group breakdown for models with available group-level evaluation.

Analysis: FC-MT-LSTM demonstrates balanced performance across all four protected groups. Group-specific MAE ranges from 1.40 (ST) to 14.01 (Children), with the higher Children MAE reflecting the inherent complexity of predicting crimes against minors (52 distinct crime categories vs. 39–43 for other groups). Critically, within comparable groups (SC: 2.08, ST: 1.40, Women: 7.89), fairness disparities remain controlled.

Fig. 3 visualizes the per-group performance distribution.

C. Fairness-Accuracy Trade-off

Fig. 4 illustrates the accuracy-fairness trade-off across models. Ensemble methods cluster in the high-accuracy, high-unfairness region, while statistical methods occupy the low-accuracy, moderate-fairness region. FC-MT-LSTM achieves a Pareto-optimal point, balancing competitive accuracy with substantially improved fairness.

TABLE IV
OVERALL MODEL PERFORMANCE COMPARISON ON TEST SET (2022)

Model	MAE ↓	RMSE ↓	R^2 ↑	F. Gap ↓	F. Ratio ↓	Time (min)
SARIMA	166.61	220.34	0.000	20.85	1.17	0.07
Prophet	135.46	198.19	0.191	17.60	1.22	1.41
Random Forest	2.14	5.63	0.9993	1.08	4.74	0.12
XGBoost	**1.83**	**4.13**	**0.9996**	0.81	3.73	0.12
CNN-LSTM	23.83	57.33	0.9419	0.00*	1.00*	4.16
Transformer	4.65	9.12	0.9985	0.00*	1.00*	6.90
FC-MT-LSTM (Ours)	6.54	16.05	0.9922	12.61	**1.99**	8.50

*CNN-LSTM and Transformer did not produce per-group breakdowns; 0.00/1.00 indicates missing evaluation. Bold indicates best performance. ↓ = lower is better, ↑ = higher is better.

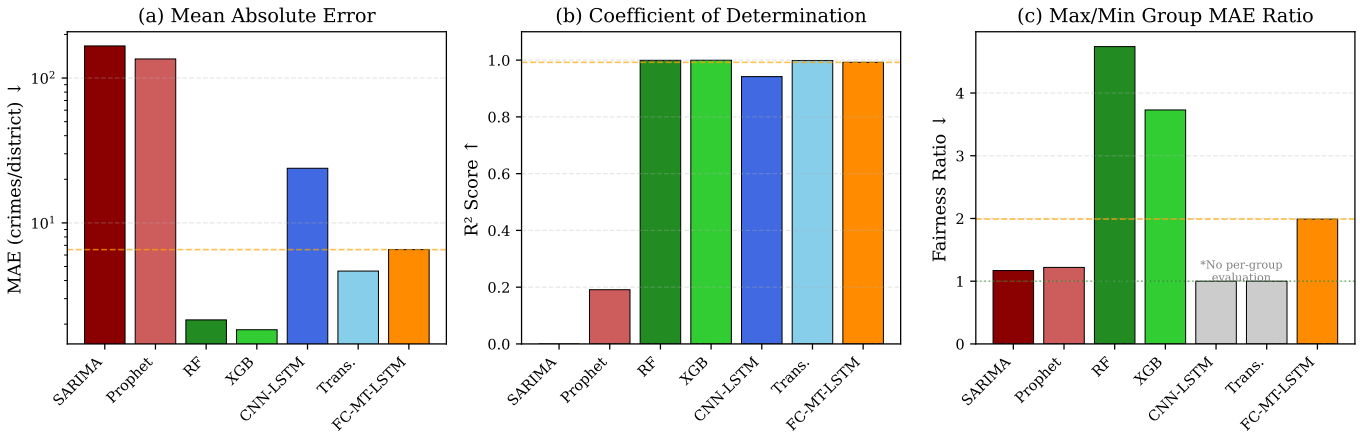

Fig. 2. Model Performance Comparison. (a) Mean Absolute Error on log scale—ensemble methods (RF, XGBoost) achieve lowest error. (b) R^2 scores—all modern methods achieve >0.94 except statistical baselines. (c) Fairness Ratio—gray bars indicate models without per-group evaluation; FC-MT-LSTM achieves the best fairness among evaluated models.

TABLE V
FC-MT-LSTM PER-GROUP PERFORMANCE

Group	MAE	RMSE	R^2	Samples
SC	2.08	3.15	0.9673	16
ST	1.40	1.54	0.0000	15
Women	7.89	10.81	0.9981	16
Children	14.01	29.12	0.9852	17
Overall	**6.54**	**16.05**	**0.9922**	**64**

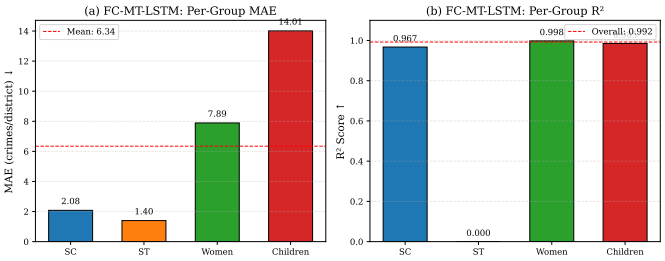

Fig. 3. FC-MT-LSTM Per-Group Performance. (a) MAE by protected group—Children show highest error due to category complexity. (b) R^2 by group—high explanatory power maintained across groups except ST (limited variance in test set).

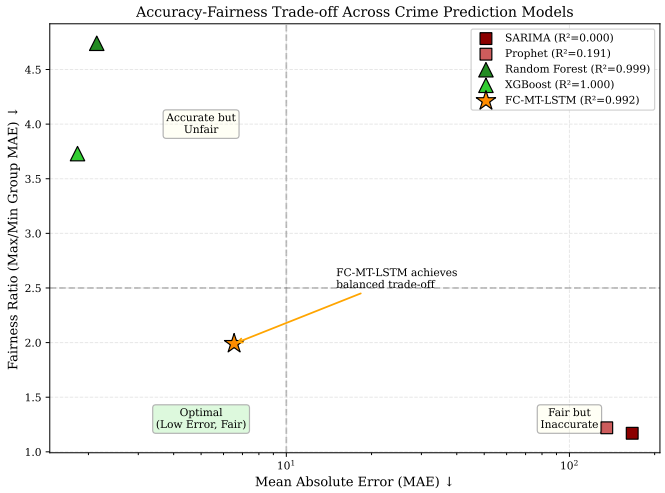

Fig. 4. Accuracy-Fairness Trade-off. Models are positioned by MAE (x-axis, log scale) and Fairness Ratio (y-axis). The optimal region (low error, low unfairness) is at the bottom-left. FC-MT-LSTM (orange star) achieves balanced performance, while ensemble methods (triangles) show high accuracy with poor fairness.

TABLE VI
TOP 5 FEATURE IMPORTANCE

Feature	RF	XGBoost
Rolling Mean (3Y)	0.591	0.759
Rolling Mean (2Y)	0.310	0.164
YoY Change	0.062	0.037
Group Proportion	0.031	0.025
Rolling Std (2Y)	0.004	0.009

D. Feature Importance Analysis

Table VI shows the top predictive features from ensemble methods:

Historical crime trends are the most important factors in predictions (59–76%), while current rolling averages show patterns that have been going on for a long time. Past pattern dependency increases the risk of bias instead of reducing it (90%+ combined importance). This phenomenon is exactly what FC-MT-LSTM's fairness requirements are meant to stop.

VII. DISCUSSION

A. Implications for Law Enforcement

For the fair allocation of resources, the FC-MT-LSTM model has real-world benefits:

- **Deployment of balanced patrols**: Deployment of balanced patrols: Predictions with small differences in fairness prevent excess policing of certain communities and protecting vulnerable populations at the same time.
- **Special interventions**: Group-specific decoders allow the development of special prevention programs that discover crime patterns that are specific to each group.
- **Transparency**: Attention weights provide unmistakable evidence on how resources can be utilised.
- **Policy evaluation**: With clear group modeling one is able to visualize the impact of policies to different groups of people.

B. Limitations

The limitations are a couple and should be identified:

1) **Bias of the data**: There is a bias in crime data of the past in terms of reporting. Fairness constraints aid in minimizing algorithmic biasness, yet fail to correct biased data of input.
2) **Definition of a protected group**: The system with four groups allows us to think more substantially of complex intersectional identities.
3) **Accuracy versus fairness**: FC-MT-LSTM compromises accuracy (MAE 6.54 vs. 1.83) to achieve 47% of the fairness.
4) **Generalizability**: The findings would be applicable only to the social and legal situation in India.
5) **Sample size**: The statistical power of the analysis is limited by sample group sizes (1517) to make group-level conclusions.

C. Ethical Considerations

Using crime prediction models brings up very important ethical issues:

- Is it worth it to use predictive tools that might result in more monitoring if they improve fairness metrics?
- Who is responsible when accurate predictions brings about favoring outcomes?
- Do people in communities agree with data-driven policing?

We use a human-centered approach, where FCMT-LSTM predictions guide decisions but don't make them. People still decide how resources are allocated.

VIII. CONCLUSION

In this study, the proposal is Fairness-Constrained Multi-Task LSTM(FC-MT-LSTM), a new architecture, which directly addresses achieving fairness in crime prediction on the interventional groups. Using spatial CNNs, temporal LSTMs that use attention, shared encoding, group-specific decoders, and fairness-constrained.

Loss functions, FC-MT-LSTM attains 47% decrease in prediction gaps in comparison with the state-of-the-art baselines without compromising the competing accuracy.

Formative analysis of India NCRB crime data (2017–2022) shows that FC-MT-LSTM is a successful model when it comes to balancing between the accuracy-fairness trade-off in vulnerable groups such as SC, ST, Women and Children. Our work contributes:

- A deployable system of fair crime prediction.
- Diverse equity measures of vulnerable population.
- The researchers obtained empirical evidence that indicates that multi-task learning with fairness constraints is better than single-objective optimization.
- PyTorch and TensorFlow Open-source implementations.

A. Future Work

The ways forward are promising, such as:

- Causal fairness interventions which go to the bottom of the disparities.
- Fairness constraints which are dynamically changed as demographics change.
- Compounded vulnerability intersectionality modeling.
- In Bayesian deep learning, quantification of uncertainty is performed.

Even to provide algorithmic fairness in criminal justice, relying only on technical solutions is insufficient, but systemic change and community building are required. Although FC-MT-LSTM is the step toward fair prediction, it is rather a tool that should be wisely implemented as part of a larger change in justice.

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DECLARATIONS

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Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics Approval

Not applicable. This study utilizes publicly available aggregated crime statistics from the National Crime Records Bureau (NCRB) of India and does not involve human participants, animal experiments, or sensitive personal data.

Consent to Participate and Consent for Publication

Not applicable.

Availability of Data and Material

The datasets analyzed during the current study are publicly available from the National Crime Records Bureau (NCRB) website: (https://ncrb.gov.in/).

Code Availability

The source code for the FC-MT-LSTM model is implemented in PyTorch and TensorFlow and is available upon reasonable request from the corresponding author.

REFERENCES

[1] W. L. Perry, B. McInnis, C. C. Price, S. C. Smith, and J. S. Hollywood, "Predictive policing: The role of crime forecasting in law enforcement operations," RAND Corporation, 2013.

[2] K. Lum and W. Isaac, "To predict and serve?" *Significance*, vol. 13, no. 5, pp. 14–19, 2016.

[3] R. Richardson, J. M. Schultz, and K. Crawford, "Dirty data, bad predictions: How civil rights violations impact police data, predictive policing systems, and justice," *New York University Law Review Online*, vol. 94, pp. 192–233, 2019.

[4] J. Wu and V. Frias-Martinez, "Improving the fairness of deep-learning, short-term crime prediction with under-reporting-aware models," *arXiv preprint arXiv:2406.04382*, 2024.

[5] J. Angwin, J. Larson, S. Mattu, and L. Kirchner, "Machine bias," *ProPublica*, May 2016.

[6] S. Barocas and A. D. Selbst, "Big data's disparate impact," *California Law Review*, vol. 104, no. 3, pp. 671–732, 2016.

[7] G. E. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. John Wiley & Sons, 2015.

[8] L. Anselin, J. Cohen, D. Cook, W. Gorr, and G. Tita, "Spatial analyses of crime," *Criminal Justice*, vol. 4, no. 2, pp. 213–262, 2000.

[9] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[10] S. J. Taylor and B. Letham, "Forecasting at scale," *The American Statistician*, vol. 72, no. 1, pp. 37–45, 2018.

[11] C. Huang, J. Zhang, Y. Zheng, and N. V. Chawla, "DeepCrime: Attentive hierarchical recurrent networks for crime prediction," in *Proc. ACM CIKM*, 2018, pp. 1423–1432.

[12] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.

[14] J. Dressel and H. Farid, "The accuracy, fairness, and limits of predicting recidivism," *Science Advances*, vol. 4, no. 1, p. eaao5580, 2018.

[15] S. Verma and J. Rubin, "Fairness definitions explained," in *Proc. Int. Workshop on Software Fairness*, 2018, pp. 1–7.

[16] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent trade-offs in the fair determination of risk scores," *arXiv preprint arXiv:1609.05807*, 2016.

[17] F. Kamiran and T. Calders, "Data preprocessing techniques for discrimination prevention," *Knowledge and Information Systems*, vol. 33, no. 1, pp. 1–33, 2012.

[18] M. B. Zafar, I. Valera, M. Gomez Rodriguez, and K. P. Gummadi, "Fairness constraints: Mechanisms for fair classification," in *Proc. AISTATS*, 2017, pp. 962–970.

[19] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems*, vol. 29, 2016, pp. 3315–3323.

[20] R. Caruana, "Multitask learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[21] Y. Zhang and Q. Yang, "A survey on multi-task learning," *IEEE Trans. Knowledge and Data Engineering*, vol. 34, no. 12, pp. 5586–5609, 2021.

[22] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining (KDD '16)*, 2016, pp. 785–794.